

→ MSE : → for normal random variable

$$MSE_{\hat{\theta}} = \text{Var}(u) + \text{Bias}^2$$

$$\text{Var}(u) = [E(\hat{\theta}^2)] - [E(\hat{\theta})]^2 = E[\bar{x}^2] - \mu^2$$

$$\text{Bias} = [E(\hat{\theta}) - \theta] \rightarrow \bar{x} - \mu$$

$E(\hat{\theta}) = \text{mean of estimator}$

→ MSE for different distributions:

Bernoulli:

$$\text{Bias} = 0, E(x) = p = E(\hat{p})$$

$$MSE(\hat{p}) = \text{Var}(\hat{p}) = p(1-p)$$

$p \rightarrow \text{probability}$

Binomial: $E[\hat{p}] = E\left[\frac{x}{n}\right] = \frac{np}{n} = p$

$$MSE(\hat{\theta}) = \frac{p(1-p)}{n}; \text{Bias} = 0$$

Poisson:

$$\text{Bias} = 0$$
$$MSE(\hat{\mu}) = \frac{\mu}{n} = \frac{\lambda}{n}$$

Normal:

$$E(\hat{\mu}) = \mu \quad [\text{bias} = 0]$$
$$\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n}$$

$$\Rightarrow MSE(\hat{\mu}) = \frac{\sigma^2}{n}$$

Mean Squared Error (MSE) for Probability Distributions

Introduction to Mean Squared Error (MSE)

Mean Squared Error (MSE) is a fundamental measure of the quality of an estimator. It quantifies the average squared difference between an estimator and the true parameter value being estimated.

General MSE Formula

For an estimator $\hat{\theta}$ of parameter θ :

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

This can be decomposed into two components:

$$MSE(\hat{\theta}) = \text{Bias}^2(\hat{\theta}) + \text{Var}(\hat{\theta})$$

Where:

- Bias: $\text{Bias}(\hat{\theta}) = E[\hat{\theta}] - \theta$
- Variance: $\text{Var}(\hat{\theta}) = E[(\hat{\theta} - E[\hat{\theta}])^2]$

1. Bernoulli Distribution

Distribution Properties

- Parameter: p (probability of success)
- PMF: $P(X = x) = p^x (1 - p)^{1-x}$ for $x \in \{0, 1\}$
- Mean: $\mu = p$
- Variance: $\sigma^2 = p(1 - p)$

MSE for Parameter Estimation

Estimating p using sample proportion $\hat{p} = \frac{X_1 + X_2 + \dots + X_n}{n}$

- Bias: $\text{Bias}(\hat{p}) = E[\hat{p}] - p = p - p = 0$ (unbiased)
- Variance: $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$
- MSE: $MSE(\hat{p}) = \frac{p(1-p)}{n}$

Key Properties:

- MSE is maximized when $p = 0.5$
- MSE decreases as sample size n increases
- For $n = 1$, MSE equals the population variance

2. Binomial Distribution

Distribution Properties

- Parameters: n (number of trials), p (probability of success)
- PMF: $P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$
- Mean: $\mu = np$
- Variance: $\sigma^2 = np(1 - p)$

MSE for Parameter Estimation

Estimating p using $\hat{p} = \frac{X}{n}$ where $X \sim \text{Binomial}(n, p)$

- Bias: $\text{Bias}(\hat{p}) = 0$ (unbiased)
- Variance: $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$
- MSE: $MSE(\hat{p}) = \frac{p(1-p)}{n}$

For multiple independent samples: If we have m independent binomial samples each of size n :

$$MSE(\hat{p}) = \frac{p(1-p)}{mn}$$

3. Poisson Distribution

Distribution Properties

- Parameter: λ (rate parameter)
- PMF: $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$
- Mean: $\mu = \lambda$
- Variance: $\sigma^2 = \lambda$

MSE for Parameter Estimation

Estimating λ using sample mean $\hat{\lambda} = \bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$

- Bias: $\text{Bias}(\hat{\lambda}) = 0$ (unbiased)
- Variance: $\text{Var}(\hat{\lambda}) = \frac{\lambda}{n}$
- MSE: $MSE(\hat{\lambda}) = \frac{\lambda}{n}$

Alternative Estimators:

- Method of Moments: Same as sample mean
- Maximum Likelihood: Also the sample mean (same MSE)

4. Normal Distribution

Distribution Properties

- Parameters: μ (mean), σ^2 (variance)
- PDF: $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
- Mean: μ
- Variance: σ^2

MSE for Parameter Estimation

Estimating the Mean μ

Using sample mean $\hat{\mu} = \bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$

- Bias: $\text{Bias}(\hat{\mu}) = 0$ (unbiased)
- Variance: $\text{Var}(\hat{\mu}) = \frac{\sigma^2}{n}$
- MSE: $MSE(\hat{\mu}) = \frac{\sigma^2}{n}$

→ Likelihood function:

→ Bernoulli:

PMF $P(X=x) = p^x(1-p)^{1-x}$

MLE of p is:

$$\hat{p} = \bar{x}$$

→ Binomial:

$$p = \frac{x}{n}$$

MLE of p is $\hat{p} = \frac{x}{n}$
111^{ly}

$$\text{MLE of } \frac{p}{1-p} = \frac{\hat{p}}{1-\hat{p}} = \frac{x}{n-x}$$

→ Poisson:

1. Bernoulli Distribution

Distribution Properties

- Parameter: p (probability of success)

- PMF: $P(X=x) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$

- Mean: $\mu = p$

- Variance: $\sigma^2 = p(1-p)$

MSE for Parameter Estimation

Estimating p using sample proportion $\hat{p} = \frac{X_1 + X_2 + \dots + X_n}{n}$

- Bias: $\text{Bias}(\hat{p}) = E[\hat{p}] - p = p - p = 0$ (unbiased)

- Variance: $\text{Var}(\hat{p}) = \frac{p(1-p)}{n}$ for variance.

- MSE: $\text{MSE}(\hat{p}) = \frac{p(1-p)}{n}$

Key Properties:

Confidence interval

- MSE is maximized when $p = 0.5$
- MSE decreases as sample size n increases
- For $n = 1$, MSE equals the population variance

→ confidence interval

→ two-sided confidence interval :

$$\text{Sample statistic} \pm [\text{Critical value} * \text{Standard error}]$$

→ one-sided :

Lower limit :

$$\text{Sample mean} - (\text{Critical value} * \text{Standard error})$$

Upper limit :

$$\text{Sample mean} + [\text{Critical value} * \text{Standard error}]$$

→ Two sided :

$$\bar{x} \pm Z_{\alpha/2} \cdot \frac{S}{\sqrt{n}}$$

$$S = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}}$$

→ one sided :

Upper bound / Lower Interval :

$$(-\infty, \bar{x} + Z_{\alpha} \cdot \frac{S}{\sqrt{n}})$$

Lower bound / Upper Interval

$$(\bar{x} - Z_{\alpha} \cdot \frac{S}{\sqrt{n}}, \infty)$$

→ In all cases 'S' can be replaced with 'σ' if known.

These formula to be applied when sample size is large [$n \geq 30$] → from CLT

→ Width of a confidence interval :

$$\begin{aligned}\text{width} &= \text{upper limit} - \text{lower limit} \\ &= 2 \times Z_{\alpha/2} \frac{S}{\sqrt{n}}\end{aligned}$$

→ Traditional method for computing confidence interval for a proportion :

$$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\hat{p} = \frac{\text{No. of Successes}}{n}$$

The method cannot be used unless the sample contains at least 10 successes and 10 failures.

→ Agresti-Coull method for proportion :
[for any sample size 'n']

$$\tilde{p} \pm Z_{\alpha/2} \cdot \sqrt{\frac{\tilde{p}(1-\tilde{p})}{\tilde{n}}}$$

$$\tilde{n} = n + 4$$

$$\tilde{p} = \frac{x + 2}{\tilde{n}}$$

→ Large-Sample point Estimators [to Estimate parameter] [one-sided interval]

To estimate one of four population parameters when the sample sizes are large, use the following point estimators with the appropriate margins of error.

Parameter	Point Estimator	Margin of Error
μ	$\bar{x}$	$\pm 1.96 \left(\frac{s}{\sqrt{n}} \right)$
p	$\hat{p} = \frac{x}{n}$	$\pm 1.96 \sqrt{\frac{\hat{p}\hat{q}}{n}}$
$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$	$\pm 1.96 \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
$p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) = \left(\frac{x_1}{n_1} - \frac{x_2}{n_2} \right)$	$\pm 1.96 \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$

→ large - sample point Estimator [for population parameter] [Two sided / bound interval] :

To estimate one of four population parameters when the sample sizes are large, use the following interval estimators.

Parameter	$(1 - \alpha)100\%$ Confidence Interval
μ	$\bar{x} \pm z_{\alpha/2} \left(\frac{s}{\sqrt{n}} \right)$
p	$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}\hat{q}}{n}}$
$\mu_1 - \mu_2$	$(\bar{x}_1 - \bar{x}_2) \pm z_{\alpha/2} \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$
$p_1 - p_2$	$(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$

→ Estimating the difference between two means

$$\bar{x} - \bar{y} \pm z_{\alpha/2} \sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}$$

σ_x can be replaced with s_x if σ is unknown.
 $[n_x \& n_y \geq 30]$

→ Students t - distribution:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

with " $n-1$ " degrees of freedom denoted by " t_{n-1} ".

→ Note: This is only used when sample data is small. If large sample data is used, the 't' curve approaches normal curve, so we just use Normal distribution.

'n' is large if $[n \geq 30]$ & when standard deviation is not known

→ Note: It only gives values for " $t > n$ " for " $t < n$ " find $t > n$ and make it negative.

→ Linear interpolation for 't'

$$y_1 = \frac{x - x_1}{x - x_2} (y_1 - y_2)$$

→ Confidence interval for a small sample:

$$\bar{x} \pm t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

$$\alpha = [100\% - \text{Confidence-level}\%]$$

→ Confidence interval between two proportions

Agresti - Caffo method:

$$p_x - p_y = \tilde{p}_x - \tilde{p}_y \pm Z_{\alpha/2} \sqrt{\frac{\tilde{p}_x(1-\tilde{p}_x)}{\tilde{n}_x} + \frac{\tilde{p}_y(1-\tilde{p}_y)}{\tilde{n}_y}}$$

$$\tilde{p}_x = \frac{(x+1)}{\tilde{n}_x} \quad \tilde{p}_y = \frac{(y+1)}{\tilde{n}_y}$$

$$\tilde{n}_x = n_x + 2 \quad \tilde{n}_y = n_y + 2$$

→ Confidence interval for true mean difference

- If n is small [$n \leq 30$]

$$\bar{D} \pm t_{n-1, \alpha/2} \frac{s_D}{\sqrt{n}}$$

- If n is large [$n > 30$]

$$\bar{D} \pm Z_{\alpha/2} \frac{s_D}{\sqrt{n}}$$